
Principal Components Analysis of Yield Curve

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Summary: This note discusses the use of principal components analysis on yield curve studies. The resulted scenario on principal components could be used for stress testing in asset/liability management.

In asset/liability management we study how assets and liabilities change along with the change in the yield curve to see if the asset matches the liability under different scenarios. We usually use a number of points to represent a yield curve, such as 10 points ranging from 3-month to 30 years. Different points on the yield curve usually move in the same direction, but by different amount. They are usually correlated with each other, which make stress testing difficult since we cannot shock each points separately from other points. Even if different points on the yield curve have independent changes we would still need to run many scenarios to capture the realistical historical change in the yield curve. For example if we use only two scenarios for each point on the yield curve we would still need to have 1024 ($= 2^5$) scenarios to represent all the possibilities. To avoid this many scenarios we usually apply some statistical approach to reduce the number of dimensions. The common approaches are principal components analysis and factor analysis.

This sort of analysis is not difficult to carry out, and is regularly done at investment banks for the term structures in many countries. For example, for the US swap curve two components capture about 92% of all observed total variations. The first of these components is a parallel shift in the term structure in which all rates move in the same direction by about the same amount. However, the 10-year rate moves by only about 80% of the change in the 3-month rate. The second component is a twisting of the term structure in which the short end and the long end move in opposite directions, pivoting at about the 2-year rate. The 10-year rate moves up by about 1/2 of the amount that the 3-month rate moves down. Similar results have been found for at least a dozen other currencies. Here we apply PCA to analyze the USD government curve change.

The way that the daily term structure changes can be decomposed by assuming that a number of common sources of uncertainty drive the changes in every rate on the term structure. This can be represented as

$$\Delta R_i = u_{1i}\epsilon_1 + u_{2i}\epsilon_2 + \dots$$

where ΔR_i is the daily change in the i th zero coupon rate in the term structure and δt is the time elapsed between observations. The random sources of uncertainty driving the term structure are the ϵ 's that are the same for all rates. The sources of uncertainty are assumed to be independent of one another and to have a variance equal to the elapsed time. The u 's are the relative importance of these sources of uncertainty for each of the rates in the term structure. These are known as factor weights. They differ from one term rate to another term rate. Thus the 3-month rate may have a large weighting on the first source of uncertainty and a small weighting on the second (i.e. u_1 for 3-month is large and u_2 for 6-month is small) while the reverse may be true for the 5-year rate.

The procedure used to estimate the factor weights is known as principal component analysis. It starts with the covariance matrix for the zero-coupon rate of difference maturities. This is generated from daily zero-coupon yield curves. The change in the rates of difference maturities are computed from day to day and used to estimate the covariance matrix of the day-to-day changes, Ω .

The model underlying principal component analysis is

$$\Delta \mathbf{R} = A\epsilon$$

where $\Delta \mathbf{R}$ is an n by 1 vector whose elements are ΔR for n different values of i , A is an n by n factor matrix that is to be determined, and ϵ is an n by 1 vector of independent factors with variance Δt , the time between observations. Note that

$$E(\Delta R \Delta R') = \Omega = AA' \Delta t$$

From this we can see that the factor weights are given by

$$A = V\Lambda^{\frac{1}{2}}$$

where V is the matrix whose columns are the eigenvectors of $\Omega/\Delta t$ and $\Lambda^{\frac{1}{2}}$ is a matrix with the square roots of the eigenvalues of $\Lambda^{\frac{1}{2}}$ on the diagonal. If the elements of $\Lambda^{\frac{1}{2}}$ are ordered from largest to smallest and the columns of V are similarly ordered, then the first column of A contains the weights for the most significant factor and the last column contains the weights for the least significant factor. The sum of the squares of the elements in each column of A is the contribution of that factor to the total variance. By using only the first m columns of A we have an m -factor model that approximates the true covariance matrix. In most cases $m = 2$ or 3 is adequate to capture the majority of the variation of the term structure.

A sample result generated for the US treasury curve is in the attached spreadsheet. Weekly changes for 3- and 6-month, and 1-, 2-, 3-, 5-, 7-, 10- and 30-year rates are computed from January 1, 1993 to March 31, 2000. The covariance matrix of the changes is computed and the principal components are generated using the statistical package of S-Plus. This was done using the actual weekly changes measured in percentage and using the change measure as a proportion of the current rates. The contents in the column labeled "Comp. 1" in the factor loading matrix are the values of u_1 above, those in the column labeled "Comp. 2" are the values of u_2 .

The first component has a factor loading all positive which corresponds to a parallel shift in the curve. The first component explains 88% of the total variation in the yield curve. The second component has negative factor loadings for short term and positive factor loading for long term, which corresponds to a twisting of the term structure. Using the first two components would explain 97% total variation of term structure. The third component has a positive factor for 3-month term, negative factor loadings for terms ranging from 6-month to 5-year, and positive loadings thereafter, which corresponds to a tilting of the yield curve. Using the first three components would explain 99% of the total variation of the term structure.

Then we could independently shock each of the three components by its two standard deviation change and translate the shock into the original 10 point on the yield curve. Each of the three components are increased or decreased by two standard deviations, which

corresponds to an “UP” or “Down” scenario. There are altogether 8 scenarios which have only one more than the current New York 7 scenarios. The only further step we need to do is to translate our yield curve scenarios into the matrix format scenarios used in TAS. We suggest to use forward rates.

//We provide a simple R code here using eigenvalue decomposition approach to PCA

```
function(x){  
  
  # x is the data set with original observation changes  
  # each column is a variable and each row is an observation  
  
  label.term <- colnames(x)  
  
  #Perform principal component analysis using singular value decomposition  
  
  x.pca <- princomp(x)  
  
  # Obtain first three factor loading vectors  
  x.loadings <- x.pca$loadings[, 1:3]  
  
  # prepare graph legends  
  legend.loadings <- c("First Principal Component", "Second Principal Component",  
    "Third Principal Component")  
  
  #Plot factor loadings  
  par(xaxt = "n")  
  matplot(x.loadings, type = "l", lwd = 3, lty = 1, xlab = "Term", ylab = "Factor  
    Loadings")  
  legend(3, max(x.loadings), legend = legend.loadings, lty = c(1, 1, 1), col  
    =c("black", "red", "green"))  
  par(xaxt = "s")  
  # add X label  
  axis(1, 1:length(label.term), label.term)
```

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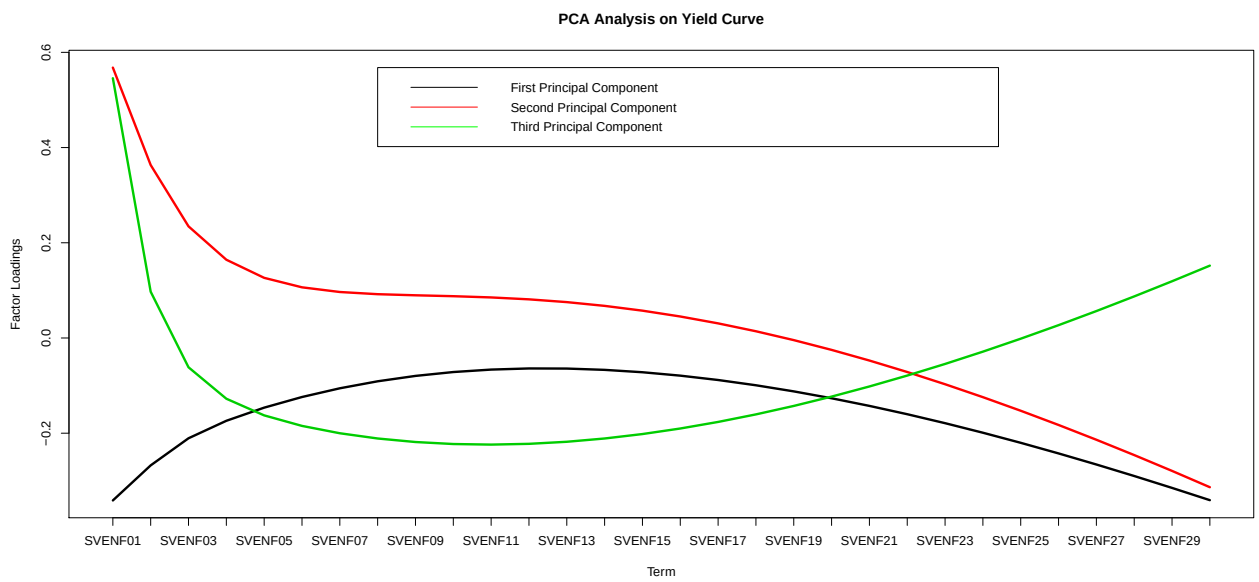


Figure 1: Yield Curve PCA